

Wipro PBL - Handling Exceptions Assignments

1. Messages table and named exception handlers

```
CREATE TABLE MESSAGES (RESULT VARCHAR2(1000));

ACCEPT VSAL NUMBER PROMPT 'Enter Salary: '
DECLARE
    VNAME EMP.ENAME%TYPE;
BEGIN
    SELECT ENAME INTO VNAME FROM EMP WHERE SAL=&VSAL;
    INSERT INTO MESSAGES VALUES (VNAME||'/'||&VSAL);
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        INSERT INTO MESSAGES VALUES('No Employee with that salary');
    WHEN TOO_MANY_ROWS THEN
        INSERT INTO MESSAGES VALUES('More than one employee with salary
'||&VSAL);
    WHEN OTHERS THEN
        INSERT INTO MESSAGES VALUES('Other Error');
END;
/
```

2. Insert employee (unnamed system exception)

```
ACCEPT E NUMBER
ACCEPT N CHAR
ACCEPT S NUMBER
BEGIN
    INSERT INTO EMP (EMPNO, ENAME, SAL) VALUES (&E, '&N', &S);
EXCEPTION
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE (SQLERRM);
END;
/
```

3. Delete employee using user-defined exception

```
ACCEPT E NUMBER
DECLARE
    INVALID_EMPNO EXCEPTION;
BEGIN
    DELETE FROM EMP WHERE EMPNO=&E;
    IF SQL%ROWCOUNT=0 THEN
        RAISE INVALID_EMPNO;
    END IF;
EXCEPTION
    WHEN INVALID_EMPNO THEN
```

```

        DBMS_OUTPUT.PUT_LINE('Employee not found');
END;
/

```

4. Stock transaction

```

CREATE TABLE STOCK (PNO NUMBER, PNAME VARCHAR2 (30), RATE NUMBER, TR_QTY
NUMBER);

```

```

ACCEPT PNO NUMBER
ACCEPT T CHAR
ACCEPT Q NUMBER
DECLARE
    PNF EXCEPTION;
BEGIN
    IF '&T'='R' THEN
        UPDATE STOCK SET TR_QTY=TR_QTY+&Q WHERE PNO=&PNO;
    ELSIF '&T'='I' THEN
        UPDATE STOCK SET TR_QTY=TR_QTY-&Q WHERE PNO=&PNO;
    ELSE
        RAISE PNF;
    END IF;
    IF SQL%ROWCOUNT=0 THEN
        RAISE PNF;
    END IF;
EXCEPTION
    WHEN PNF THEN
        DBMS_OUTPUT.PUT_LINE('Invalid Product/Transaction');
END;
/

```